MUHAMMAD ALI AAMIR

Full-Stack Developer | AI/ML Engineer | Data Science Student

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PROFESSIONAL SUMMARY

Data Science student at FAST-NUCES Lahore with hands-on experience in AI engineering, software engineering, and full-stack development. Builds machine learning and AI systems including RAG pipelines, fine-tuned language and document models, computer vision applications, and agentic workflows, alongside production web applications and serverless systems. Experienced across Python, JavaScript/TypeScript, React, Flask, FastAPI, Cloudflare, AWS, and modern data and AI tooling.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, C++, C, SQL, x86 Assembly

AI / Machine Learning: Machine Learning, Deep Learning, Generative AI, Agentic AI, RAG Pipelines, LLM Fine-Tuning, Computer Vision, NLP, TensorFlow, Keras, scikit-learn, Hugging Face, LangChain, OpenCV, YOLOv8

Web / Backend: React.js, Next.js, Vite, Node.js, Express.js, Flask, FastAPI, REST APIs, Tailwind CSS, HTML5, CSS3

Cloud / DevOps: AWS, Google Cloud, Microsoft Azure, Vercel, Cloudflare Workers, D1, Durable Objects, AI Gateway, Docker, CI/CD, Git, GitHub

Data / Databases: Pandas, NumPy, Apache Spark, Power BI, Feature Engineering, Predictive Analytics, Data Visualization, Streamlit, Redis, MongoDB, PostgreSQL, Microsoft SQL Server

EXPERIENCE

Software Engineering Intern — Exaverse | May 2026 – Present | Remote

- Build serverless applications on Cloudflare Workers, Durable Objects, D1, and AI Gateway.
- Integrate generative AI and agentic AI workflows into edge-deployed production architecture, including RAG-powered features with deterministic validation gates and human-in-the-loop review.

AI Engineering Intern — Vertekx | Jun 2026 – Aug 2026 | Remote

- Worked on an open-ended financial-document AI task where the main challenge was to find a practical approach under strict cost, security, accuracy, and inference-time constraints.
- Independently researched OCR engines, document parsers, model architectures, and related tooling; read documentation, built small tests, and compared approaches on sample data before selecting the components that fit the requirements.
- Explored and fine-tuned multiple model approaches for the required document-classification results, including a LayoutLMv3-based model across 17 audit classes, while keeping the pipeline fully offline and avoiding external LLM/API calls.
- Built the final inference and validation layer with FastAPI and led Phase 1 of the project from research and experimentation through the working implementation.

Founder & Developer — Kryzect | 2024 – Present | Lahore, Pakistan (Remote)

- Founded and lead a software engineering and AI company delivering full-stack web, iOS/Android, and AI product engineering solutions for B2B and B2C clients.
- Architect MERN, Flask, and Next.js systems from requirements through deployment on AWS, Cloudflare, Vercel, and Render.
- Deliver Figma-to-React builds that consistently reach 95+ Lighthouse scores and manage delivery from project scoping through client handoff.

SELECTED PROJECTS

IncidentIQ — AI-Powered Incident Postmortem Automation | Cloudflare Workers, RAG, Agentic AI

- Designed a multi-agent system with a central orchestrator coordinating retrieval, analysis, and validation sub-agents for RAG-grounded root-cause analysis.
- Added a mandatory human-in-the-loop review gate with deterministic validation before generated output is surfaced.

GitHub | Live Demo

Criclytics — Cricket Analytics Platform | Python, Flask, scikit-learn, JavaScript

- Built context-aware ML models for win probability and player performance analysis using venue, opposition, pitch, and form data.
- Processed thousands of historical matches through custom feature engineering and added Redis caching and lazy-loaded UI components for responsive live probability updates.

GitHub | Live Demo

Financial Document Classifier — Fine-Tuned DistilBERT | Hugging Face, FastAPI, Python

- Fine-tuned DistilBERT to classify financial documents across 17 audit categories entirely offline, with zero external LLM API calls.
- Built OCR and image-extraction pipelines supporting native/scanned PDF, DOCX, CSV, and XLS/XLSX inputs, served through a FastAPI inference layer.

GitHub

Fashion Image Classifier — CNN & Random Forest | TensorFlow, Keras, scikit-learn, Streamlit, OpenCV

- Developed a dual-model clothing classifier across 8 categories, reaching 92.25% accuracy with a custom CNN and 87.56% with a Random Forest baseline.

- Applied data augmentation and stratified sampling to address class imbalance and built a Streamlit interface for real-time prediction.
  GitHub

TravelBuddy — AI-Powered Travel Platform | React, Node.js, Express, SQL, AI/ML

- Built a full-stack social travel application with an ML-driven itinerary generator, buddy matching, and multi-currency expense sharing backed by a SQL database.
  GitHub

Flappy Bird & Bubble Shooter — 16-bit Assembly Games | x86 Assembly, NASM, VGA Graphics, DOSBox

- Wrote two arcade games in 16-bit x86 Assembly with fixed-point physics and direct VGA memory rendering.
- Implemented custom collision detection, pseudo-random number generation, and interrupt-driven keyboard input without a floating-point unit.
  GitHub

EDUCATION

BS Data Science — FAST-NUCES, Lahore | Aug 2024 – Jun 2028 (Expected)
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Linear Algebra, Statistics, Object-Oriented Programming, Operating Systems, Computer Organization & Assembly Language

LANGUAGES

English (Fluent) | Urdu (Native) | Punjabi (Native)